

Question Number	Error	Line Number	Error Type	Error Correction
1	int choice	23	Syntax Error	Missing semicolon at the end of the declaration: int choice;
2	default	59	Syntax Error	A switch label needs a colon. Use: default:
3	if(index > 5)	85	Logical Error (Incorrect Bounds Check)	students[] has size 5 (valid indices 0-4), but the check allows index to reach 5 before flagging "Database Full", causing an out-of-bounds write. Check before writing and use: if(index >= 5) { printf("Database Full\n"); return; }
4	for(i=0;i<=totalStudents;i++)	95	Logical Error (Off-by-One)	Using <= accesses one element past the last valid student. Use: for(i=0;i<totalStudents;i++)
5	return total / 0;	119	Runtime Error (Division by Zero)	Should divide by the number of students, not the literal 0. Use: return totalStudents > 0 ? total / totalStudents : 0;
6	int highest = students[0].marks;	124	Type Mismatch Error	marks is a float; storing it in an int silently truncates the decimal part. Declare as: float highest = students[0].marks;
7	int lowest = students[0].marks;	126	Type Mismatch Error	marks is a float; storing it in an int silently truncates the decimal part. Declare as: float lowest = students[0].marks;
8	totalStudent	148	Semantic Error (Undeclared Identifier)	Typo: the actual variable is "totalStudents". Correct it to: totalStudents
9	printGrade();	150	Semantic Error (Undeclared/Undefined Function)	printGrade() is never declared or defined anywhere in the file. Add a prototype and a matching function definition, or remove the call.
10	calculateRank();	152	Semantic Error (Undeclared/Undefined Function)	calculateRank() is never declared or defined anywhere in the file. Add a prototype and a matching function definition, or remove the call.
11	s.rollNo = "101";	159	Type Mismatch Error	rollNo is an int; cannot assign a string literal. Use: s.rollNo = 101;
12	s.marks = "90";	163	Type Mismatch Error	marks is a float; cannot assign a string literal. Use: s.marks = 90;
13	students[roll].marks = 100;	176	Runtime Error (Unvalidated Array Index)	roll comes directly from user input and is never checked against array bounds (0 to totalStudents-1) before use. Validate first: if(roll >= 0 && roll < totalStudents) students[roll].marks = 100;
14	students[roll] = NULL;	185	Compile Error (Type Mismatch)	students[roll] is a struct, not a pointer; NULL cannot be assigned to it. Also roll is never bounds-checked. Reset the struct instead, e.g. memset(&students[roll], 0, sizeof(struct Student)); after validating roll.
15	if(name == students[0].name)	194	Logical Error	This compares array addresses, not the string contents. Use: if(strcmp(name, students[0].name) == 0)
16	avg = calculateAverage();	204	Type Mismatch Error	calculateAverage() returns a float; storing it in an int (avg) truncates the decimal part. Declare as: float avg;
17	fp = fopen("student.txt", "r");	218	Logical Error (Wrong File Mode)	The file is opened in read-only mode ("r") but then written to with fprintf. Open it for writing instead: fp = fopen("student.txt", "w"); and check fp != NULL before using it.
18	fp = fopen("missing.txt", "r");	229	Runtime Error (Missing NULL Check)	If the file does not exist, fopen returns NULL, and the following fscanf(fp,...) call will crash. Add a check: if(fp == NULL) { printf("File not found\n"); return; }